D · SEED

R&D BRIEF — DEGRADABLE MOTE

SEED-DROP SEASONAL SENSOR — RESEARCH PROGRAM

AF-D-101

REV 0 · 2026-11-12

RESEARCH — NOT A PRODUCT

THE IDEA

- A ~20 mm mote of printed electronics on biodegradable substrate — seed-dropped over a fuel break or burn scar.
- It sprouts into a temporary sensor skin for fire season, reports through cluster heads, and dissolves.
- Zero e-waste, no retrieval, no grid to build — sensing that ENDS by design.
- This is a materials/transient-electronics program — fundable R&D, not a product. Every earlier rung must succeed first.

WHY THIS MATTERS

- Fire season is seasonal — permanent sensors waste 11 months of life.
- Degradable motes match the threat's rhythm: deploy for the season, vanish after.
- Billions-scale is materials R&D, not drone fleet — the mesh math supports it (duty cycle + clustering).
- Precedent: DARPA/NSF computational-dust programs prove the research is real.

RESEARCH TARGETS

Target	Goal
Mass	<1 g per mote (cluster head <5 g)
Cost	<$1/unit at volume (printed, not assembled)
Lifetime	One fire season (~6 months) then degrade
Degradation	Substrate gone in <12 months, non-toxic residue
Comms	Backscatter or recycled LoRa — rare TX, local clustering
Power	Printed battery + PV/TEG harvest — µW average
Sensing	Temp/RH/VOC minimum; optical smoke later

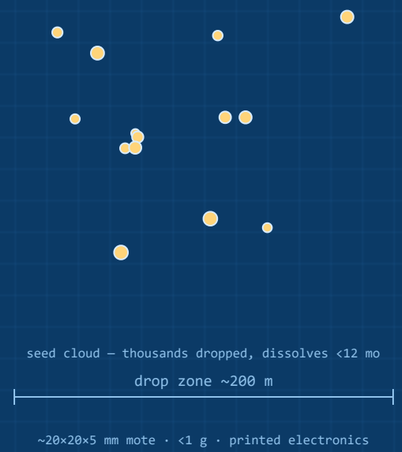
MATERIALS PATHWAY

Component	Candidate materials / approach
Substrate	Cellulose paper, silk fibroin, or transient polymer
Battery	Printed Zn/MnO_2 battery, printed sensors, organic TFT
Energy	Small PV cell, thermoelectric, or RF harvest
Comms	Backscatter (no TX power) or ultra-low-power LoRa wake-up
Activation	Soluble coating — germinates sensor on moisture

HONEST STATUS

- TRL ~2 — concept + literature. No ArcticChain mote exists yet.
- This sheet is a research brief, not a buildable BOM — deliberately.
- The swarm architecture (roles, perch, mesh) is the same — only the node shrinks.
- Do NOT promise motes to funders until Gate 2 demo exists.

FORM FACTOR TARGET



ROADMAP GATES

- Gate 1: university transient-electronics partner identified (SR&ED eligible).
- Gate 2: lab demo — printed mote senses temp + reports 10 m through cluster head.
- Gate 3: field trial — 100 motes over 1 ha, one season, recoverable data.
- Gate 4: degradable variant — residue analysis confirms <12 month breakdown.
- Gate 5: scale — seed-drop deployment tied to Anchor/Brick infrastructure.

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